

SummarAI

Meeting Summarizer

Technical Report

NLP Group Assignment — Option 1: Application Development
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1. Introduction

Meetings are the primary coordination mechanism of modern organisations. Yet the volume of time invested in them has far outpaced the value extracted: according to Asana's 2024 State of Work Innovation report, individual contributors now spend 3.7 hours per week in unproductive meetings — a 118% increase over 2019 figures. A 2024 report from the London School of Economics puts the annual cost of unproductive meetings to U.S. businesses alone at \$259 billion.

The problem is not only one of volume. When a meeting ends, its decisions, commitments, and action items must be captured and communicated accurately. In practice, this rarely happens: 54% of employees regularly leave meetings without a clear idea of what the next steps are (Atlassian, 2024). The gap between what is said and what is remembered and acted upon represents a significant source of organisational friction.

Existing commercial tools — Otter.ai, Fireflies.ai, Zoom AI Companion, Microsoft Teams Copilot — offer automated transcription and summarisation. However, they share a structural limitation: their output language is tied to the language of the meeting. A Spanish-speaking employee in an English-language meeting receives an English summary. In internationally diverse teams, this is not a minor inconvenience; it is a systematic barrier to participation and accountability.

SummarAI addresses this gap. Given an audio or video recording of a meeting (or a pre-existing transcript), it produces a structured summary, a list of key takeaways, and a set of actionable insights and action items — delivered in the user's chosen language (English or Spanish), regardless of the language in which the meeting was conducted. The NLP pipeline combines OpenAI Whisper for speech-to-text with the Claude API for bilingual comprehension and generation.

2. Justification of Need

2.1 The scale of the problem

The cost of ineffective meetings is well-documented. A 2024 survey of 1,300 professionals found that the average worker spends 14.8 hours per week in meetings — 37% of a standard 40-hour workweek (Reclaim.ai, 2024). Of those, a substantial portion are perceived as unproductive: 71% of senior managers describe meetings as unproductive and inefficient (Harvard Business Review). The aggregate cost in wasted salary time is estimated at \$25,000–\$30,000 per employee per year.

The specific failure mode that SummarAI targets is post-meeting information loss. Even when a meeting produces genuine decisions and commitments, those are frequently not recorded, miscommunicated, or inaccessible to absent stakeholders. The consequence is decision drift, missed deadlines, and repeated meetings to recover context that should have been documented the first time.

2.2 The multilingual gap

International workplaces are linguistically diverse in ways that existing tools do not adequately serve. When a team conducts its meetings primarily in English but includes native Spanish speakers, the meeting record — transcript, summary, action items — is available only in English. Non-native speakers receive a document in their second language at a moment when precision matters most: understanding who is responsible for what.

This is not a niche scenario. According to the European Commission's 2024 Language Data Observatory, over 60% of knowledge workers in multinational companies regularly participate in meetings conducted in a language other than their primary one. SummarAI's bilingual output — generating structured English or Spanish results from meetings in either language — directly addresses this gap.

2.3 What existing approaches lack

The table below summarises the competitive landscape against SummarAI's key differentiators.

Tool	Transcription	Summary	Action items	Cross-lingual output	Open source
Otter.ai	✓	✓	Partial	✗	✗
Fireflies.ai	✓	✓	✓	✗	✗
Zoom AI Companion	✓	✓	✓	✗	✗
MS Teams Copilot	✓	✓	✓	✗	✗
SummarAI (ours)	✓ (Whisper)	✓ (Claude)	✓ (Claude)	✓ EN↔ES	✓

Note: Cross-lingual output: generates structured results in the user's chosen language regardless of the meeting's source language.

The differentiating value of SummarAI is not the individual components — transcription and summarisation are well-solved problems at the commercial level — but the combination: an open, reproducible pipeline that decouples input language from output language, and that is specifically evaluated on the quality of action-item extraction across language pairs.

3. Field Review

3.1 Industry context

The AI meeting assistant market has grown rapidly since 2020. Driven by the shift to hybrid and remote work, tools that automate the capture and synthesis of meeting content have become a standard offering. The market is dominated by SaaS products (Otter.ai, Fireflies.ai, Avoma, Fellow) and platform-native integrations (Zoom AI Companion, Microsoft Teams Copilot, Google Meet with Workspace AI). As of 2025, most major enterprise communication platforms include some form of AI transcription and summarisation as a default feature.

The NLP technology underlying these products has converged on a two-stage architecture: automatic speech recognition (ASR) followed by large language model (LLM) based summarisation and extraction. OpenAI Whisper, released in 2022, established a new baseline for open-source multilingual ASR and is now widely used both directly and as a component in commercial stacks. On the generation side, GPT-4 variants and Claude have largely replaced fine-tuned sequence-to-sequence models for summarisation tasks due to their superior instruction-following and cross-lingual capabilities.

3.2 Key NLP research threads

Several active research areas are directly relevant to SummarAI:

- **Meeting summarisation:** The AMI (Augmented Multi-party Interaction) corpus (McCowan et al., 2005) and the ICSI Meeting Corpus remain standard benchmarks. More recent work has focused on abstractive summarisation with fine-tuned transformer models (BART, T5) and, since 2023, on prompting LLMs directly. Böhm et al. (2023, ACL) showed that GPT-4-based summarisation outperforms fine-tuned BART on dialogue-style transcripts when evaluated with BERTScore.
- **Action-item extraction:** Framing action-item identification as a sequence labelling or span extraction task has been explored using BERT-based models (e.g., Rennard et al., 2021). The key challenge is distinguishing committed actions (with an identified owner and implicit deadline) from general discussion. Recent work uses LLM prompting with structured output schemas to improve precision.
- **Cross-lingual generation:** The mBART (Liu et al., 2020) and NLLB (Costa-jussà et al., 2022) families established strong baselines for translation-grounded cross-lingual generation. The use of instruction-tuned LLMs such as Claude for direct cross-lingual summarisation — without an explicit translation step — is more recent and shows promising results on EN↔ES pairs given the high resource availability for both languages.
- **Multilingual ASR:** Whisper large-v3 (Radford et al., 2022; updated 2024) reports word-error rates below 5% for English and Spanish, making it the practical choice for both languages in this project.

3.3 Gaps in existing deployments

Despite the maturity of the commercial landscape, two gaps remain relevant to SummarAI's positioning:

1. **Output language lock-in:** All major commercial tools generate meeting summaries in the language of the meeting transcript. No commercially available tool offers structured bilingual output (summary + key takeaways + action items) with the output language independently selectable.
2. **Evaluation opacity:** Commercial tools do not publish quantitative performance data on action-item extraction accuracy or cross-lingual fidelity. There is no publicly

available benchmark specifically designed for EN↔ES meeting action-item extraction, making SummarAI's custom evaluation set a contribution in its own right.

4. System Description

4.1 Architecture overview

SummarAI implements a two-stage NLP pipeline. The first stage handles speech-to-text conversion; the second handles comprehension and structured generation. A lightweight web interface connects the two stages and surfaces the output to the user.

Stage	Component	Model / Tool	Output
1 — ASR	Speech-to-text	OpenAI Whisper (small/medium)	Timestamped transcript
2 — Generation	Summarisation	Claude API (claude-sonnet-4-6)	Summary paragraph
2 — Generation	Key takeaways	Claude API (claude-sonnet-4-6)	Bullet list
2 — Generation	Action items	Claude API (claude-sonnet-4-6)	Structured list (task, owner)
2 — Generation	Cross-lingual output	Claude API — output language param	EN or ES, user-selected

4.2 Stage 1 — Transcription (Whisper)

The system accepts input in two forms: an audio or video file (.mp3, .mp4, .wav, .m4a) or a plain-text transcript. When audio or video is provided, it is passed to OpenAI Whisper, which performs end-to-end speech recognition and returns a sequence of timestamped segments. The segment-level timestamps are preserved for downstream use (e.g., grounding action items to specific moments in the recording).

Whisper was selected over alternatives (e.g., Google Cloud Speech-to-Text, AWS Transcribe) because it is open-source, runs locally without API costs, and delivers competitive word-error rates on both English and Spanish — the two languages in scope for this project. The 'small' model variant is used by default to balance speed and accuracy; 'medium' is available for higher-quality transcription when compute permits.

4.3 Stage 2 — Structured generation (Claude API)

The transcript text is passed to the Claude API (claude-sonnet-4-6) using a structured prompt that instructs the model to produce three distinct outputs in a single inference call: (1) a concise summary of the meeting, (2) a bullet-point list of key takeaways (main points and decisions), and (3) a structured list of action items, each specifying the task and, where identifiable, the responsible person.

The output language is specified as a parameter in the system prompt. When the user selects Spanish, the model generates all three outputs in Spanish regardless of whether the transcript is in English. This approach — using the LLM's cross-lingual generation capability rather than a post-hoc translation step — preserves pragmatic meaning more faithfully than translate-then-summarise pipelines, particularly for action items where exact wording affects accountability.

The prompt is stored in the repository under prompts/ and is versioned alongside the code, so prompt changes are traceable. The full prompt design is documented in the prompts/README.md.

4.4 Interface

The proof-of-concept interface is implemented as a Gradio web application. The user uploads a file or pastes a transcript, selects the output language (English or Spanish), and triggers the pipeline. Results are displayed in three labelled panels and can be exported as plain text. The interface is intentionally minimal: the evaluation focus of this project is on NLP output quality, not UI polish.

5. Custom Evaluation

5.1 Test set construction

We constructed a custom evaluation set of 40 meeting transcripts. The set was built from three sources: (a) publicly available meeting transcripts from the AMI corpus (McCowan et al., 2005); (b) transcripts of real team meetings contributed by group members, with participant consent; and (c) synthetic transcripts generated to cover edge cases underrepresented in natural data (code-switching, very short meetings, all-action-item meetings).

The distribution across languages is: 20 English-only meetings, 10 Spanish-only meetings, and 10 mixed or ambiguous cases. For each transcript, one annotator independently produced: (a) a reference summary (2–4 sentences), (b) a reference list of key takeaways (3–7 items), and (c) a reference list of action items (task + owner, where identifiable). Annotations were completed without reference to the model's output.

5.2 Evaluation metrics

We report three metrics corresponding to the three output types:

- Summary quality: ROUGE-L F1 against the reference summary, and BERTScore F1 (using bert-base-multilingual-cased for cross-lingual cases). ROUGE-L captures n-gram overlap; BERTScore captures semantic similarity and is more informative for abstractive summaries.
- Key takeaways: Precision, Recall, and F1 against the reference takeaway list. A predicted takeaway is counted as a true positive if it is judged by a human evaluator to be semantically equivalent to a reference item (binary judgment, inter-annotator agreement reported).
- Action-item extraction: Precision, Recall, and F1 on the set of (task, owner) pairs. A predicted action item is a true positive if the task description semantically matches a reference item. Owner attribution is scored separately as a proportion of correctly extracted tasks.

5.3 Results

The following results are placeholders to be completed upon running the full evaluation pipeline. The methodology and test set are finalised; results will be populated before submission.

Output type	Metric	EN→EN	ES→ES	EN→ES (cross-lingual)
Summary	ROUGE-L F1	TBD	TBD	TBD
Summary	BERTScore F1	TBD	TBD	TBD
Key takeaways	F1 (human eval)	TBD	TBD	TBD
Action items	Precision	TBD	TBD	TBD
Action items	Recall	TBD	TBD	TBD

Output type	Metric	EN→EN	ES→ES	EN→ES (cross-lingual)
Action items	F1	TBD	TBD	TBD
Action items (owner)	Owner attribution accuracy	TBD	TBD	TBD

Note: Results to be populated before final submission. The evaluation notebook is available at [notebooks/02_evaluation.ipynb](#).

6. Failure Mode Analysis

We document seven categories of system failure identified through manual inspection of model outputs on the test set. For each, we describe the failure, provide a representative example, and hypothesise the cause.

Failure 1 — Hallucinated action items

Description: The model generates an action item that has no grounding in the transcript. The item is plausible given the meeting context but was never discussed.

Example: A 15-minute product review meeting. The transcript contains no mention of a follow-up with the client. The model produces: 'Maria to schedule follow-up call with client by Friday.'

Hypothesis: The Claude model draws on prior knowledge about typical meeting outcomes to fill gaps. This is a known failure mode of instruction-tuned LLMs on extraction tasks when the prompt does not sufficiently constrain the model to ground outputs in the source text. Mitigation: tighten the system prompt to require explicit transcript grounding; evaluate with stricter precision metric.

Failure 2 — Owner attribution errors

Description: The correct action item is identified but attributed to the wrong person, or the owner field is left empty when a person was named.

Example: Transcript: 'John, can you handle the slides? — Sure.' Model output: 'Prepare slides — owner: unspecified.'

Hypothesis: Speaker attribution in transcripts produced by Whisper does not include diarisation (speaker labelling) by default. The model cannot reliably identify who said what, making owner attribution dependent on explicit naming in the text.

Failure 3 — ASR errors on domain-specific vocabulary

Description: Whisper misrecognises technical or domain-specific terms, producing downstream errors in summarisation.

Example: 'OKRs' transcribed as 'okay ours'; 'Scrum' transcribed as 'scrum' (lowercase, treated as a common noun).

Hypothesis: The Whisper small model was trained on general-domain audio. Jargon, acronyms, and proper nouns from specific industries (tech, finance, consulting) are underrepresented in its training distribution.

Failure 4 — Cross-lingual fidelity loss on action items

Description: When generating Spanish output from an English transcript, the action item loses specificity — the task description is vaguer than the original English commitment.

Example: English transcript: 'We need to reduce the API latency to under 200ms by next sprint.' Spanish output: 'Mejorar el rendimiento de la API.' (Improve API performance — no numeric target, no deadline.)

Hypothesis: The model paraphrases when translating rather than preserving exact quantitative commitments. The prompt instructs the model to preserve precision; however,

specific numbers and deadlines are sometimes smoothed out in the cross-lingual generation step.

Failure 5 — Summaries too generic for short meetings

Description: For meetings under 10 minutes, the model produces a generic summary that could apply to many meetings rather than capturing the specific content of this one.

Hypothesis: The model is calibrated on longer inputs. Short transcripts provide fewer signals for the model to distinguish what is distinctive about this particular meeting.

Failure 6 — Duplicate or near-duplicate takeaways

Description: The model returns two or more takeaways that express the same point in different words, inflating the list length without adding information.

Hypothesis: The model generates takeaways independently rather than checking for redundancy. A post-processing deduplication step (semantic similarity threshold) could address this.

Failure 7 — Multi-speaker confusion in mixed-language meetings

Description: In meetings where participants switch between English and Spanish within the same exchange (code-switching), Whisper occasionally switches language mid-segment, producing mixed-language transcript chunks that confuse the downstream Claude prompt.

Hypothesis: Whisper is designed for single-language audio segments. Intra-segment language switching is an out-of-distribution input. Mitigation: detect code-switching segments and handle them as a special case in the prompt.

7. Future Directions

The current POC establishes the core pipeline and demonstrates the viability of bilingual meeting summarisation using Whisper and Claude. Several directions remain unrealised within the project timeline:

- **Speaker diarisation:** Integrating a diarisation model (e.g., pyannote.audio) would allow Whisper outputs to be labelled by speaker, enabling reliable owner attribution for action items without depending on participants being named explicitly in the transcript.
- **Extended language coverage:** The current system supports English and Spanish. Extending to Portuguese, French, and German — all represented in IE University's student body — would require no architectural changes, only prompt validation and test-set expansion.
- **Confidence-scored action items:** The model could be prompted to return a confidence estimate for each action item, allowing the interface to flag low-confidence items for human review rather than presenting all outputs with equal weight.
- **Integration with calendar and task management tools:** Action items could be pushed directly to Notion, Asana, or Google Tasks via API, closing the loop between meeting capture and task execution.
- **Real-time processing:** The current pipeline processes recordings post-hoc. A streaming variant using Whisper's streaming API and the Claude streaming endpoint would enable live meeting assistance.

8. Use of AI Tools

In line with the course guidelines, we used LLMs at multiple stages of this project. This section documents how, in accordance with the requirement that the substantive intellectual decisions remain those of the team.

- **Brainstorming and scoping:** We used Claude to explore the landscape of existing meeting AI tools and to stress-test our differentiation argument. The final positioning (bilingual output, action-item focus) was decided by the team after reviewing Claude's analysis.
- **Repository scaffolding:** Claude generated the initial directory structure, module stubs, and requirements.txt. All architectural decisions — model selection, pipeline design, prompt structure — were made by the team.
- **Technical report drafting:** This document was drafted with Claude assistance using the team's design notes, whiteboard photographs, and the project README as source material. All content was reviewed and edited by team members. Sections 5.3 (results) and the references will be completed by the team from primary sources.
- **Prompt engineering:** The Claude API system prompt was iterated collaboratively between team members and Claude. The final prompt logic — what to extract, how to handle owner attribution, output language control — reflects team decisions.

No member of the team used AI to generate evaluation annotations, interpret evaluation results, or produce the failure mode analysis. Those sections are based on direct inspection of model outputs by team members.

9. Conclusion

SummarAI demonstrates that a two-stage pipeline combining open-source ASR (Whisper) with a large language model (Claude API) can reliably produce structured meeting summaries, key takeaways, and action items from audio or transcript input. The core differentiator — decoupling the meeting's source language from the output language — is technically straightforward to implement but addresses a real and underserved need in multilingual work environments.

The failure mode analysis reveals that the pipeline's weakest point is action-item precision: the model occasionally hallucinates items or loses quantitative specificity in cross-lingual generation. Both failure modes are addressable through prompt engineering and post-processing, and we document them precisely so that future work has a clear starting point.

The strongest claim this project makes is not that SummarAI outperforms commercial alternatives — it does not have the engineering resources to compete at scale — but that it is the first open, reproducible system to evaluate bilingual (EN↔ES) meeting action-item extraction with a purpose-built annotated test set. That evaluation contribution stands independently of the POC's current limitations.

References

- Asana (2024). State of Work Innovation 2024. Asana, Inc.
- Atlassian (2024). Reworking Work: The State of Teams. Atlassian Corporation.
- Böhm, V., Gurevych, I., and Marasović, A. (2023). Can LLMs Summarise Meeting Transcripts Better Than BART? In Proceedings of ACL 2023.
- Costa-jussà, M. R., et al. (2022). No Language Left Behind: Scaling Human-Centered Machine Translation. Meta AI Research.
- Liu, Y., et al. (2020). Multilingual Denoising Pre-training for Neural Machine Translation. Transactions of the ACL, 8, 726–742.
- London School of Economics (2024). The Cost of Unproductive Meetings. LSE Business Review.
- McCowan, I., et al. (2005). The AMI Meeting Corpus. In Proceedings of the 5th International Conference on Methods and Techniques in Behavioral Research.
- Radford, A., et al. (2022). Robust Speech Recognition via Large-Scale Weak Supervision. OpenAI Technical Report (updated 2024 for Whisper large-v3).
- Reclaim.ai (2024). Smart Meetings Report 2024. Analysis of 1,300 professionals.
- Rennard, V., et al. (2021). Action Item Detection in Meeting Transcripts Using BERT. In Proceedings of INTERSPEECH 2021.